TECHHAVEN
STUDIOS

HavenDOS

v0.8.0 Beta
“Beacon”

OPERATOR’S MANUAL

```
+-------------------------+
| 32-bit x86  .  real mode |
| to protected mode in one |
| jump, and everything     |
| after it written here.   |
+-------------------------+
```

THS-HD-0800-OM-A

October 2026

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1 About HavenDOS

HavenDOS is a 32-bit operating system for x86 machines, written from nothing by TechHaven Studios. It is not a Linux distribution, not a DOS clone and not built on anyone else’s kernel. The bootloader hands control to code in this source tree and everything after that — the memory manager, the drivers, the filesystems, the TCP/IP stack, the shell, the desktop — is in it.

It looks like 1988 on purpose and runs like it too: the whole system is a character grid, 80 columns by 25 rows, and every program in it draws with characters. What changed in this release is that the grid is no longer made of VGA text memory. It is drawn pixel by pixel into a framebuffer at 1280×800, which is why the letters are sharp on a modern screen and why the system boots on UEFI machines that have no text mode at all.



FIGURE 1-1 The system describing itself.

What this manual covers

This is the operator’s manual for HavenDOS 0.8 Beta, codename “Beacon”. It covers installing the system, what happens when it boots, every screen the setup wizard shows, the desktop, the shell and its commands, the hardware it drives, and — in section 14 — what it cannot do yet. A beta manual that only listed the good parts would waste your afternoon.

The shape of the system

Area	Lines	What is in it
kernel	1,950	boot, GDT/IDT, the PIT, physical memory, the heap, the kernel log, config
drivers	4,520	VGA and the framebuffer console, keyboard, mouse, PCI, UHCI (USB), ATA, ATAPI, VirtIO, RTL8139, e1000, RTC, speaker
fs	3,470	FAT12, FAT16, ISO 9660, a RAM filesystem, the VFS over them, the installer, the updater
net	4,150	Ethernet, ARP, IPv4, ICMP, UDP, TCP, DHCP, DNS, HTTP, TLS
shell	17,120	the shell and its commands, the desktop, the apps, the games, setup, packages
tests	3,470	437 checks that run against the real driver code on the build machine

> **NOTE**
> Line counts are approximate and are there to show the balance between the parts, not to be precise.

2 What you need

A machine

HavenDOS runs on a 32-bit or 64-bit x86 machine, real or virtual. It wants very little:

Requirement	Minimum	Notes
CPU	any 32-bit x86	it runs in protected mode and uses no instruction newer than a 486
Memory	16 MB	256 MB is what the test machines are given
Disk	15 MB	the installer refuses anything smaller
Display	any	BIOS VGA text or a GRUB framebuffer; UEFI needs the framebuffer, which it gets
Firmware	BIOS or UEFI	the ISO carries a boot image for both

A disk controller it can talk to

This is the part that decides whether an install will work, so it is worth reading before you burn anything. HavenDOS drives:

- **Legacy IDE / ATA** at the standard ports 0x1F0 and 0x170. This is what a PIIX3 chipset gives you — in QEMU, the `i440fx` machine type; in UTM, an emulated PC.
- **VirtIO block**, which is what UTM SE and most paravirtual setups offer.
- **ATAPI CD-ROM** on the same legacy ports, which is how the installer and the updater read the ISO.

> **CAUTION**
>
> HavenDOS does **not** drive AHCI or NVMe. On a Q35 or ICH9 machine the CD-ROM is behind an AHCI controller, so the system will boot and install but `update` will not find the ISO. Use an i440FX machine type if you want in-place updates.

Known-good configurations

Host	Settings
QEMU	`-M pc` (i440FX), 256 MB, IDE disk, IDE CD-ROM, `-usb`
UTM SE	x86_64, emulated PC, 512 MB, VirtIO or IDE disk, CD/DVD on IDE
Real PC	BIOS or UEFI boot, SATA set to **Legacy IDE** in firmware setup

3 Installing

Write `HavenDOS-0.8.0.iso` to a USB stick or attach it as a CD-ROM, and boot from it. The ISO is hybrid: the same image boots on a BIOS machine and on a UEFI machine without any switch.



FIGURE 3-1 The boot menu. Four entries, all from the same kernel; the version in the footer is generated at build time rather than painted into the background image.

The installer

The install ISO boots straight into the installer. There is no live desktop on the install media — the ISO exists to put the system on a disk. It runs in seven steps.

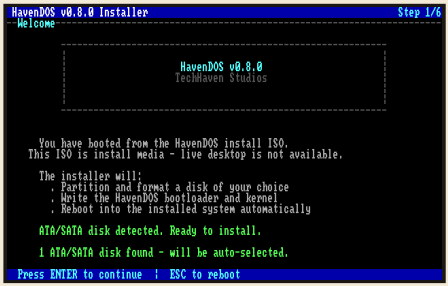


FIGURE 3-2 Step 1. The terms, which you must scroll to the end of before the accept key does anything.



FIGURE 3-3 Step 2. The privacy policy.

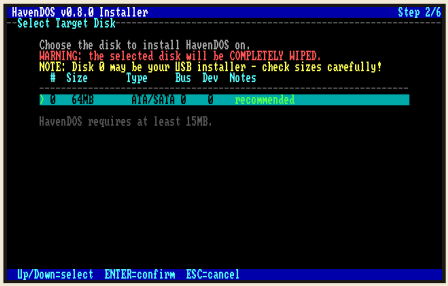


FIGURE 3-4 Disk selection, with the size of each candidate. Read the sizes: the stick you booted from is a disk too.

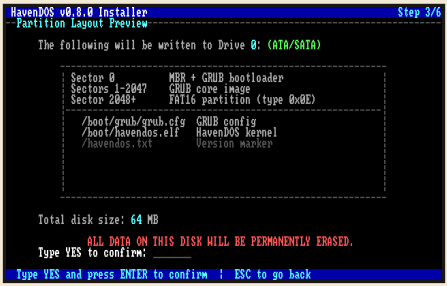


FIGURE 3-5 Exactly what will be written, shown before anything is.

> **CAUTION**
>
> The selected disk is wiped completely. The installer asks you to type `YES` rather than press a key, because a key is too easy to press by accident.

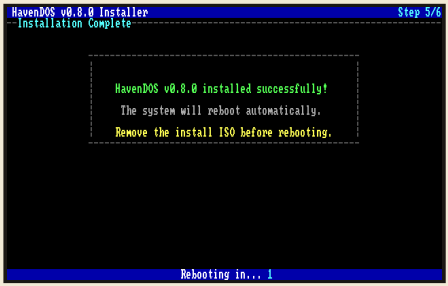


FIGURE 3-6 Partitioning, formatting FAT16, writing GRUB to the MBR and the kernel into /boot.

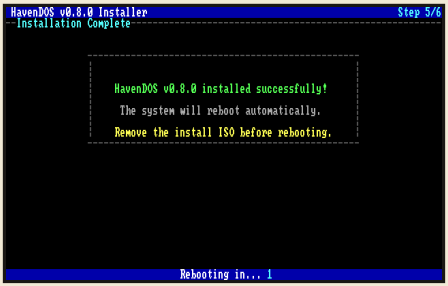


FIGURE 3-7 Done, and restarting.

The live shell

The boot menu's **Shell Mode** entry gives a shell with nothing installed. It is the quickest way to find out whether HavenDOS can see a machine before committing a disk to it: `lspci -v`, `usb -vv`, `dmesg` and `ifconfig` all work there. `install` from that shell runs the same installer with a disk list you choose from.

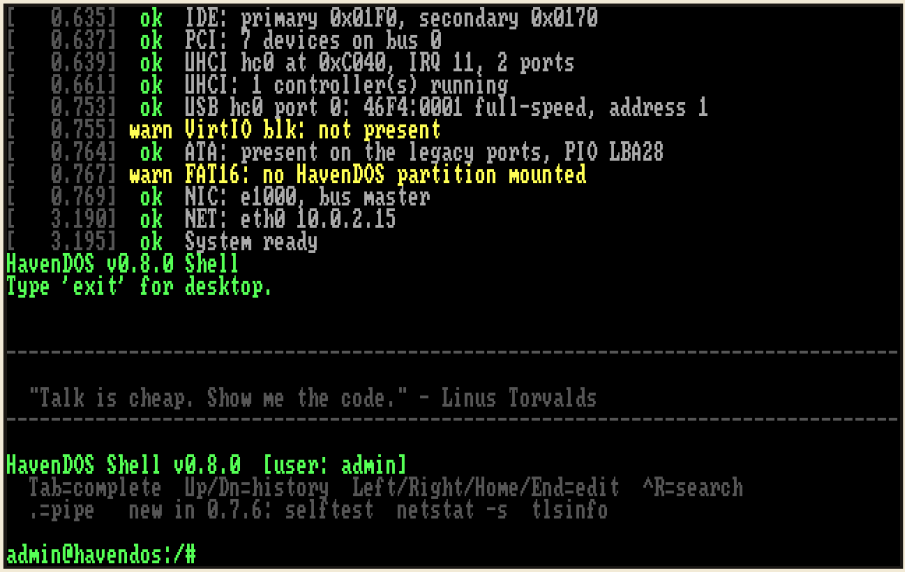


FIGURE 3-8 The ISO's live shell. The boot log is still on screen above the prompt, the way a Linux console leaves it.

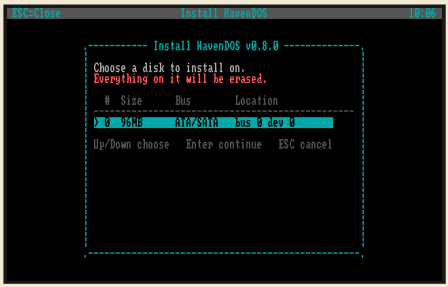


FIGURE 3-9 The installer's disk list: bus and size for each candidate.

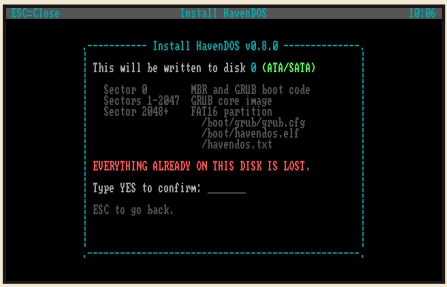


FIGURE 3-10 What will be written, and the confirmation it asks for.

What ends up on the disk

```
Sector 0          MBR + GRUB boot code
Sectors 1-2047    GRUB core image
Sector 2048+      FAT16 partition, type 0x0E
                    /boot/grub/grub.cfg
                    /boot/havendos.elf
                    /havendos.txt        version marker
```

The partition is an ordinary FAT16 volume. Any other operating system can read it, which matters when something goes wrong and you want to look at the kernel file yourself.

4 Starting up

Three things happen between power and a prompt, and all three are visible on purpose.

The boot menu

GRUB offers the desktop, shell mode, safe mode (no mouse) and a reinstall entry. It times out into the desktop after ten seconds.

The splash

A short splash with the name, version, channel and codename. It holds for about half a second and any key cuts it short. It makes no claim about your hardware, which is a change: earlier releases showed a fake power-on self test announcing a “TechHaven BIOS” that does not exist, after the real firmware had already done the real POST.

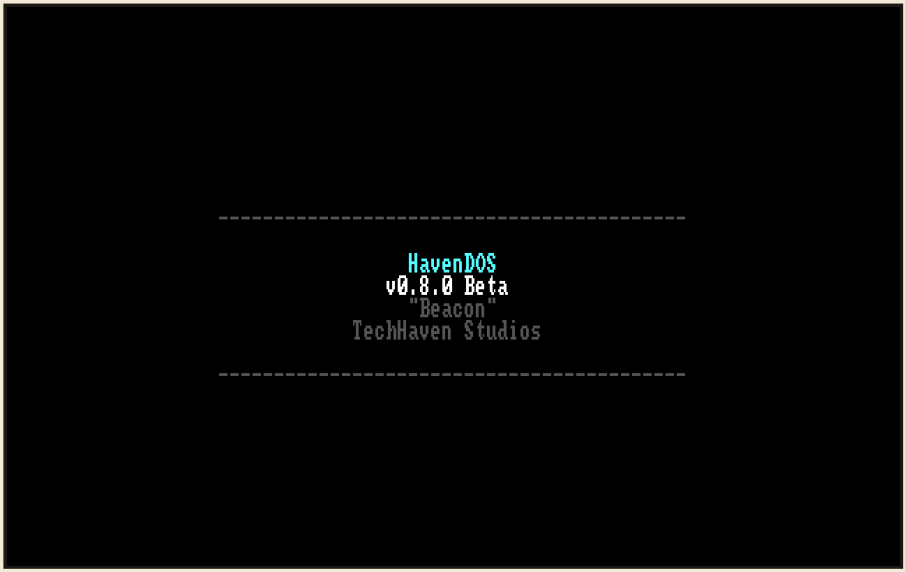


FIGURE 4-1 The splash. Name, version, channel, codename; nothing it cannot back up.

The boot log

Then each subsystem reports as it initialises, with a real timestamp in milliseconds taken from the programmable interval timer. Every line is written from inside the code that knows the number, so the log describes the machine in front of you rather than a machine someone imagined while writing string literals.

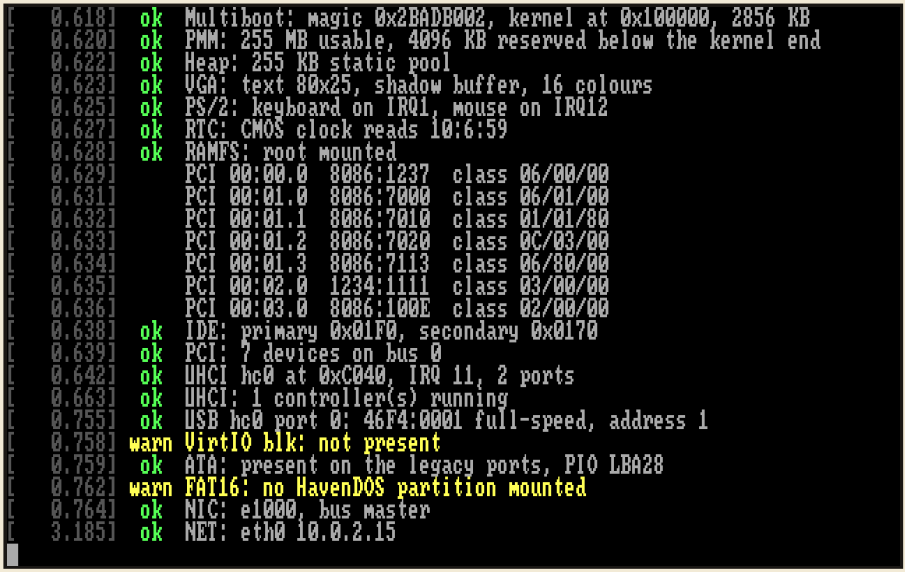


FIGURE 4-2 The verbose boot log. Each line is printed by the subsystem it describes, at the moment it happens.

```
[   0.700]  ok  GDT: 3 descriptors at 0x229000, flat 32-bit segments
[   0.704]  ok  Heap: 255 KB static pool
[   0.644]  ok  IDE: primary 0x01F0, secondary 0x0170
[   0.649]  ok  UHCI hc0 at 0xC040, IRQ 11, 2 ports
[   0.763]  ok  USB hc0 port 0: 46F4:0001 full-speed, address 1
[   0.767] warn ATA: nothing on 0x1F0/0x170 (no legacy IDE here)
[   3.184]  ok  NET: eth0 10.0.2.15
```

Echo is on from the very first line, so a kernel that dies during initialisation has already said how far it got. Add `quiet` to the kernel command line to silence it, and run `dmesg` afterwards to replay it.

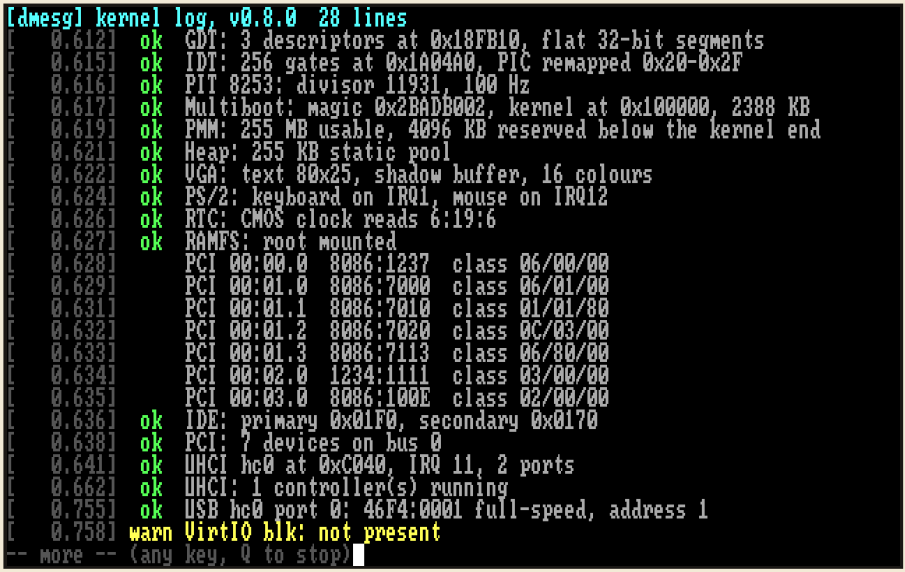


FIGURE 4-3 dmesg replays the same log, a page at a time.

> **NOTE**
>
> The log is the first place to look when hardware is not found. A line like ATA: nothing on 0x1F0/0x170 tells you immediately that the machine has no legacy IDE, which is why a later update would fail.

Boot parameters

Parameter	Effect
mode=desktop	start the desktop (the default)
mode=shell	start the shell directly
mode=safe	shell with the mouse disabled
mode=lowres	force the smaller text grid
install=1	run the installer
quiet	suppress the boot log (dmesg still has it)

Parameter	Effect
mouse=on|off	override mouse detection
gfx=text|mode13|vesa640|vesa800	the graphics mode apps use

5 First run

The first time the installed system boots it runs a setup wizard. It runs once.

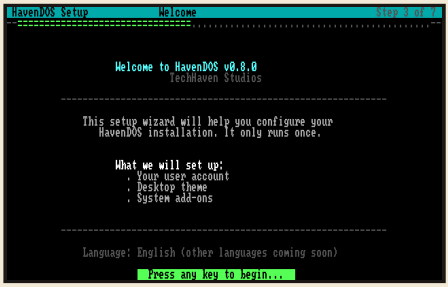


FIGURE 5-1 What setup is about to do.



FIGURE 5-2 Your account. Three characters or more; a blank password is allowed and means no password.

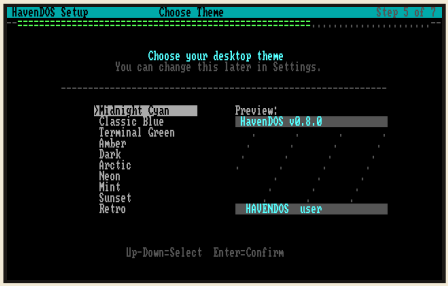


FIGURE 5-3 Twenty themes, with a live preview of the desktop in each.

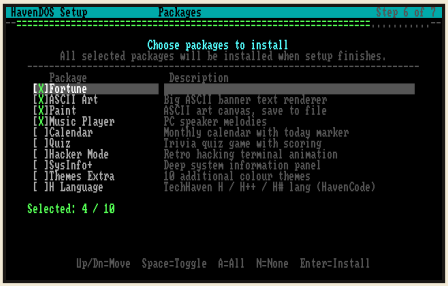


FIGURE 5-4 Optional packages. Four are selected by default.

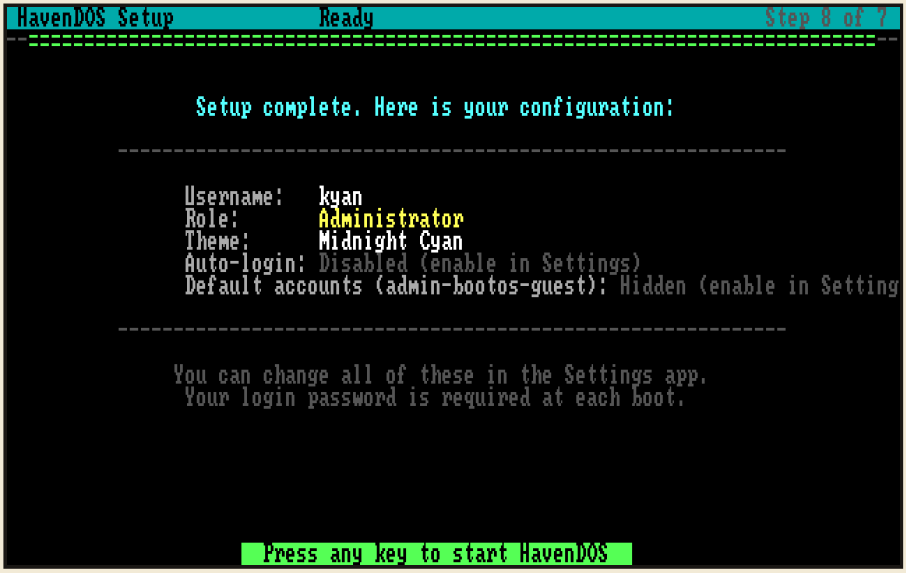


FIGURE 5-5 Setup finished.

Signing in

After setup the login manager appears. It offers a new account or one of the three built-in ones — `admin`, `bootos` and `guest` — which exist so a system you have locked yourself out of is still usable.



FIGURE 5-6 New account, or a built-in one.



FIGURE 5-7 Choosing a user.

> **CAUTION**
>
> The built-in accounts have no password until you give them one. On any machine other people can reach, set one with `passwd admin` as your first action.

6 The desktop

The desktop is an icon grid with a status bar along the bottom and a clock. Arrow keys move the selection, Enter opens, Tab goes straight to the shell and Escape opens the menu. A mouse works if one is attached.

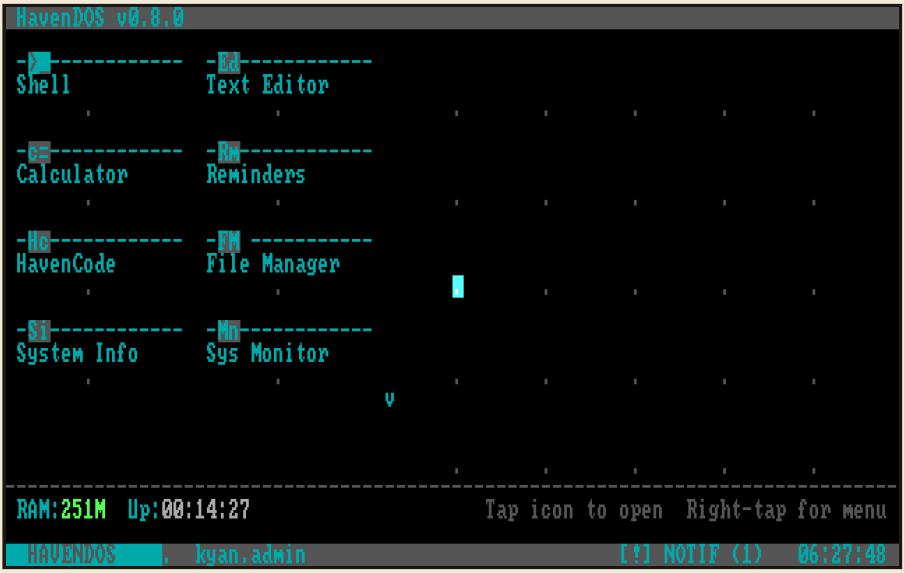


FIGURE 6-1 The desktop. Memory and uptime on the left of the status bar, the user and notifications on the right.

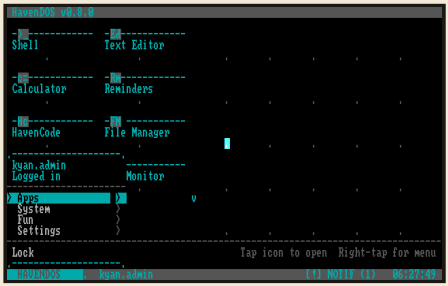


FIGURE 6-2 The menu: Apps, System, Fun and Settings, plus Lock.

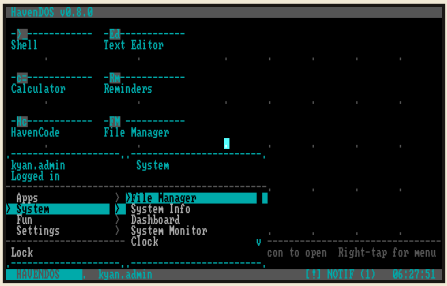


FIGURE 6-3 A submenu flies out to the right.

What is on it

Category	Programs
Apps	Shell, Text Editor, Calculator, Todo List, HavenCode, Hex Viewer, File Manager
System	System Info, Dashboard, System Monitor, Clock, Disk Info, Network Info, User Manager, Log Viewer
Fun	Snake, Tetris, Pong, Matrix, Pipes
Settings	Screensaver, Settings, Power / Reboot



FIGURE 6-4 The calculator.

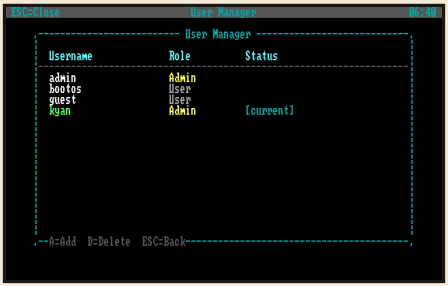


FIGURE 6-5 The user manager.

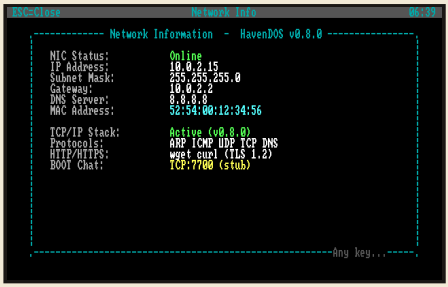


FIGURE 6-6 Network information.

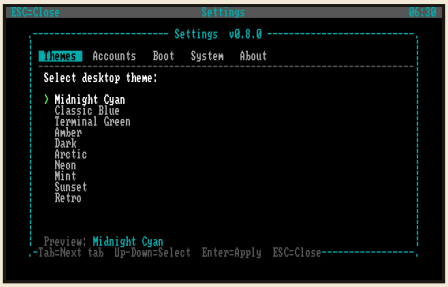


FIGURE 6-7 Settings.

> **NOTE**
>
> A program launched from the desktop clears the screen, runs, and clears it again on the way out. In releases before 0.8 the desktop stayed visible underneath, showing through wherever the program did not draw.

7 The shell

The shell is where most of the system is. It has tab completion, command history on the arrow keys, reverse search on Ctrl-R, line editing with Home and End, environment variables, aliases, redirection, pipes and scripts.

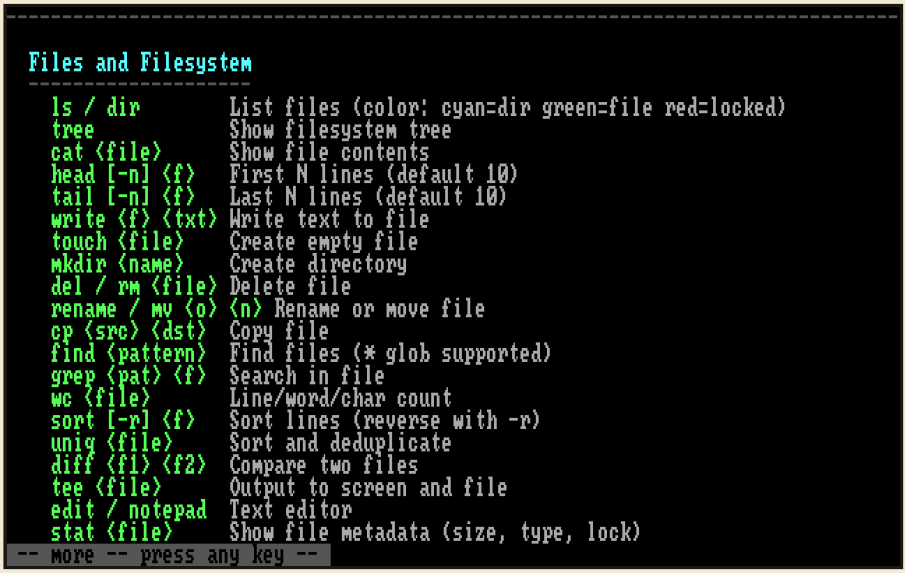


FIGURE 7-1 help lists every command, grouped by what it does.

Pipes and redirection

A full stop is the pipe character, because the vertical bar is awkward on several of the keyboard layouts HavenDOS is used with.

```
kyan@havendos:~# ls . sort > listing.txt
kyan@havendos:~# cat listing.txt . wc
kyan@havendos:~# echo $USER
kyan@havendos:~# set GREETING hello
kyan@havendos:~# echo $GREETING
```

Scripts

`run script.bsh` executes a file of shell commands. `if`, `else`, `endif`, `while`, `endwhile` and `repeat` are available, which is enough for the kind of small job a script is actually used for.

Getting help

> **NOTE**
> If a command is not on the help screen, try it anyway and try `man` on it. Tab completion knows every command the shell can run, which is the authoritative list.

`help` is the index; `man <command>` is the detail; `guide` is a built-in tour for someone who has just arrived.

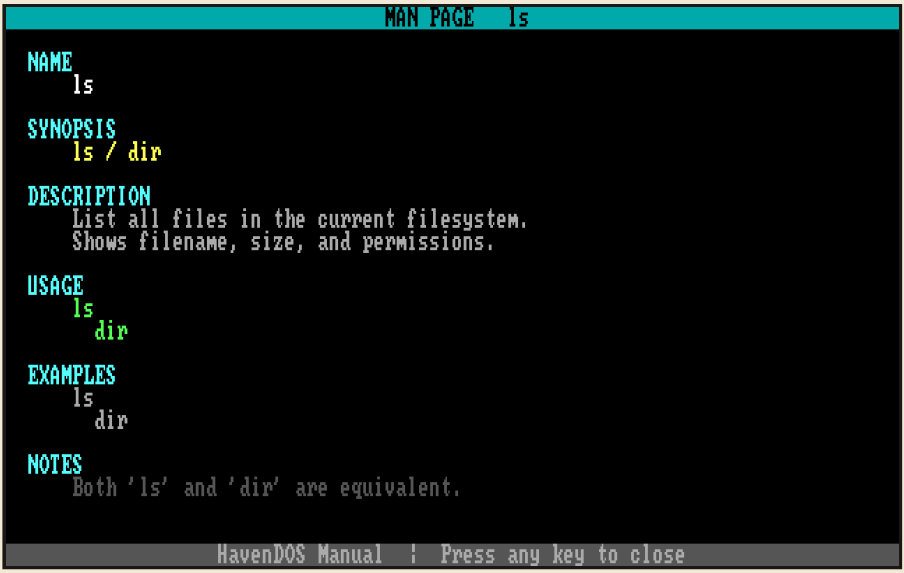


FIGURE 7-2 A manual page.

8 Command reference

Files and filesystem

Command	What it does
ls, dir	list files; cyan is a directory, green a file, red locked
tree	the filesystem as a tree
cat	show a file
head, tail	the first or last lines of a file
write	write text to a file
touch	create an empty file
mkdir	create a directory
rm, del	delete a file
mv, rename	rename or move
cp	copy
find	find files, with `*` globbing
grep	search inside a file
wc	count lines, words and characters
sort, uniq	sort lines; sort and deduplicate
diff	compare two files
stat	size, type and lock state
hex	hexdump, or `hex -s` for strings
zip, unzip	archive and extract
edit	the text editor

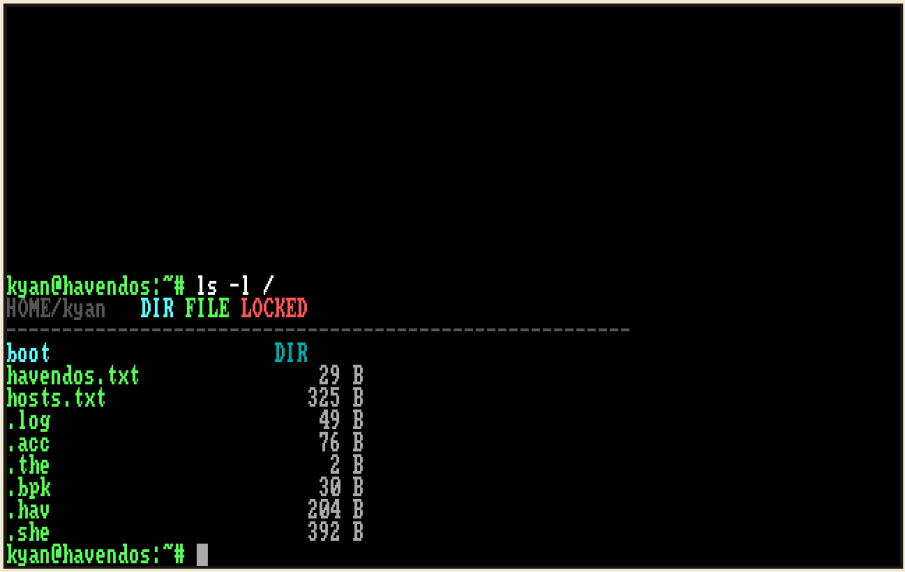


FIGURE 8-1 A directory listing.

Shell and scripting

Command	What it does
echo	print text, expanding `$VAR`
set, unset, export, env	environment variables
alias, unalias	command aliases
history	what you have typed
ps1	show or set the prompt format
run	run a .BSH script
which	does this command exist
calc	an expression calculator
repeat, seq	run something N times; a number sequence
clear, cls	clear the screen

System

Command	What it does
sysinfo	everything the kernel knows about the machine
neofetch	the same, short, with the logo
mem	physical memory and the heap
uptime	how long the machine has been up
disk	disks, partitions and the filesystem
df	free space
dmesg	replay the kernel log
ps	the process list
x86	the CPU as detected
lspci	every PCI device, decoded; `-v` for detail
usb	USB controllers and devices; `-v`, `-vv`
dashboard, htop	a live monitor
selftest	the built-in self test
ver, version	version, channel and codename
date, time	the real-time clock
reboot, shutdown	restart or power off

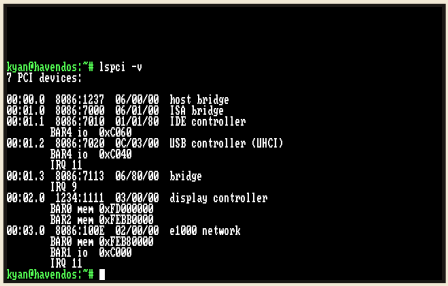


FIGURE 8-2 The PCI bus, decoded by class.



FIGURE 8-3 USB controllers, ports and what is attached.

Users and security

Command	What it does
whoami	who you are
users	every account
passwd	change a password
adduser, deluser	add or remove an account (admin only)
chmod	lock a file (admin only)
chown	change a file’s owner (admin only)
log	the audit log (admin only)
lock	lock the screen

Network

Command	What it does
ifconfig	show or set the interface configuration
dhcp	ask for an address
ping	ICMP echo

Command	What it does
netstat	connections; `-s` for per-protocol counters
nslookup, dns, host	resolve a name
wget, curl	fetch a URL over HTTP
udpsend, tcpsend, tcptest	send datagrams, or open a connection by hand
tls, tlsinfo	the TLS layer and what it negotiated
chat	the chat client

Fun and extras

Command	What it does
calendar	this month, with today marked
todo	the to-do list
bpkg	list and install the optional packages
h, h-lang	HavenCode, the TechHaven H language
snake, tetris, pong	games
matrix, pipes, stars	screensavers you can run on demand
fortune	a quote
banner, ascii-art	large letters
cowsay	an important system utility
quiz, hacker	a trivia quiz; a fake intrusion
music, play, beep	the PC speaker
paint	an ASCII canvas

9 Storage and files

HavenDOS has a VFS with three filesystems under it.

Filesystem	Used for	Access
FAT16	the installed system partition	read and write
FAT12	floppy images	read and write
ISO 9660	the install and update media	read
RAM	scratch files before an install	read and write

The installed partition is plain FAT16 with type 0x0E, which means any other system can mount it. Files carry an owner and a lock flag; `chmod` and `chown` set them and the listing shows a locked file in red.



FIGURE 9-1 What the disk layer sees.

The editor

`edit` is a full-screen editor with the usual movement keys, insert and overwrite, and page up and down.

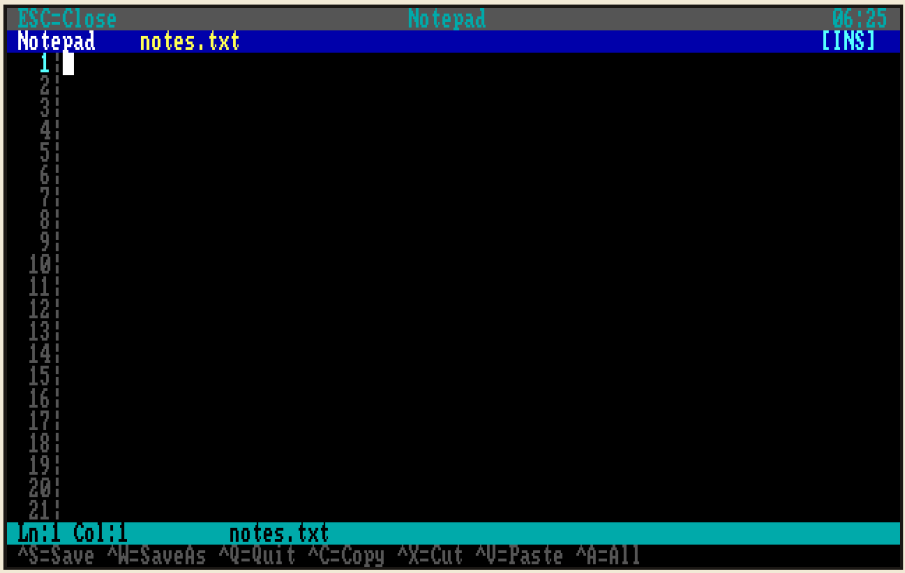


FIGURE 9-2 The editor.

10 Networking

The network stack is written from the frame up: an Ethernet driver, then ARP, IPv4, ICMP, UDP and TCP on top of it, then DHCP, DNS, HTTP and TLS on top of those. Two NICs are driven — the Realtek RTL8139 and the Intel e1000 — which between them cover almost every virtual machine.



FIGURE 10-1 The interface and its address.



FIGURE 10-2 Per-protocol counters, which are the quickest way to see where packets are being lost.

An address is obtained by DHCP at boot. `ping` goes out over the real stack, not an emulation of one, and `nslookup` resolves through a real DNS query.



FIGURE 10-3 ICMP echo.

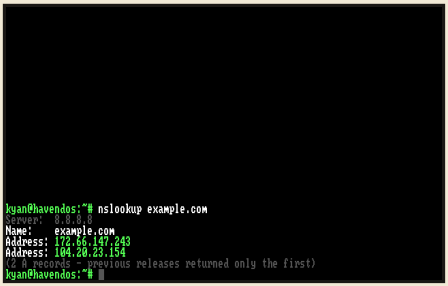


FIGURE 10-4 A DNS lookup.



FIGURE 10-5 Fetching a page over HTTP.

> **NOTE**
>
> TLS is present and negotiates, but treat it as experimental. `tlsinfo` reports what was agreed.

11 Hardware

The PCI bus

The bus is walked at boot and every device recorded with its class, vendor and device ID. Multifunction devices are enumerated properly — which sounds like housekeeping and is not. Before 0.7.20 the scan looked at function 0 only, and on a PIIX3 chipset the USB controller is function 2 of device 1. It was not undetected; it was undetectable.

USB

HavenDOS drives UHCI, which is USB 1.1. It resets a port, assigns an address and reads a device’s descriptor, and it has working control and bulk transfer primitives. Every controller on the machine is driven, not just the first: a PIIX3 has one, an ICH has three to five, and a device in any of the others used to be invisible with no error anywhere.

There is one piece of this worth understanding, because it is why USB appears to do nothing on a modern chipset. A USB 2.0 controller is an EHCI with UHCI companions sharing the same physical sockets, and which one gets a device is decided by speed: a 2.0 stick attaches to the EHCI at 480 Mbit and the companion sees an empty port. HavenDOS asks the EHCI for the ports back — the specification provides for exactly this, and it is the state the hardware resets into — and the device re-attaches at full speed to a controller this kernel can drive.

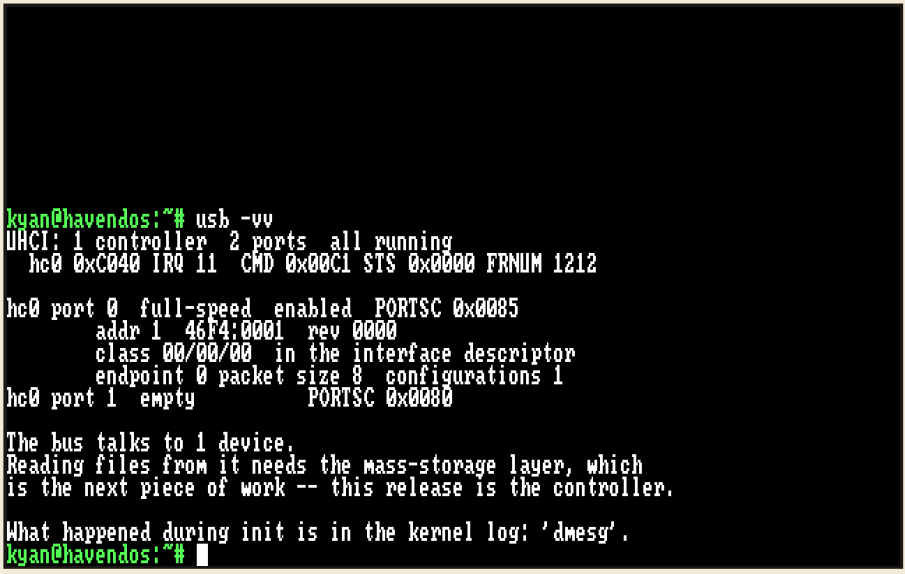


FIGURE 11-1 usb -vv: controllers, ports, and what is on them.

> **CAUTION**
>
> A USB stick is **seen**, not **read**. Mass storage needs Bulk-Only Transport and a small SCSI layer on top of the transfer primitives that now exist. That is the next piece of work, not a feature of this release.

Everything else

Device	Driver
Display	VGA text, and a framebuffer console at whatever mode GRUB provides
Keyboard	PS/2, with the full scancode set
Mouse	PS/2, used by the desktop
Disk	ATA/IDE (PIO LBA28) and VirtIO block
CD-ROM	ATAPI on the legacy ports
Network	RTL8139 and Intel e1000

Device	Driver
Clock	the CMOS real-time clock
Timer	the 8253 PIT at 100 Hz, which is also where boot timestamps come from
Sound	the PC speaker



FIGURE 11-2 The CPU, as the kernel detects it.

12 The display

Every character HavenDOS draws goes into a shadow buffer, and exactly one piece of code turns a cell in that buffer into something visible. That is why this release could change what a cell looks like without touching a single program.

On a BIOS machine with no framebuffer, a cell is written to VGA text memory at 0xB8000 and the card draws it. When GRUB provides a framebuffer — which it now does, because the kernel asks for one — a cell is drawn glyph by glyph into pixels at 1280×800. The 8×8 font is scaled twice across and four times down, which gives exactly 80×25 and is how an 8×8 font looked on a 400-line display in the first place.

This is also why HavenDOS boots on UEFI. A UEFI machine has no VGA text memory and no INT 10h; without the framebuffer console it comes up to a black screen, and with it, to a usable shell.

Changing the grid

Command	Result
res 80 25	80×25, the default, with tall glyphs
res 80 50	80×50, uniform scaling and square pixels
res 90 30	90×30 (text mode only)
res 90 60	90×60 (text mode only)

Themes

Twenty themes ship with the system, from Midnight Cyan through Amber, Terminal Green and Monochrome to Volcano. `theme` lists them and `theme <n>` switches, immediately.

13 Apps, games and packages

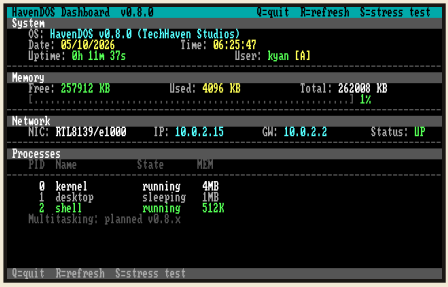


FIGURE 13-1 The live system dashboard.

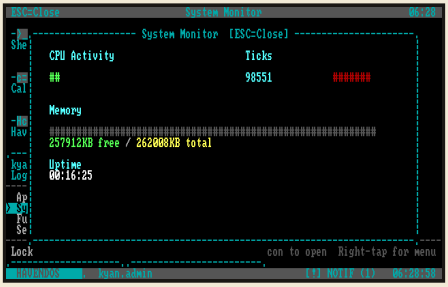


FIGURE 13-2 The system monitor.

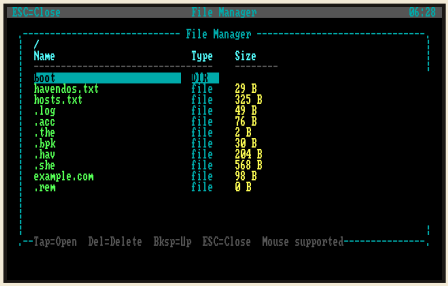


FIGURE 13-3 The file manager.

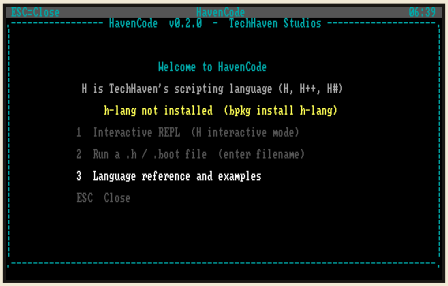


FIGURE 13-4 HavenCode, the H language environment.

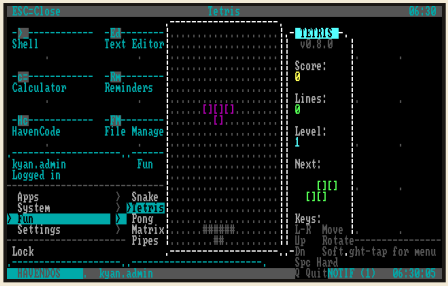


FIGURE 13-5 Tetris.



FIGURE 13-6 Pong.



FIGURE 13-7 Snake.



FIGURE 13-8 The ASCII canvas.

Packages

Ten optional packages can be installed at setup or afterwards with `bpkg`. They are part of the kernel image rather than separate binaries — the package system records which are enabled, not which are present.

Package	What you get
Fortune	a quote generator
ASCII Art	a large-letter banner renderer
Paint	an ASCII canvas you can save to a file
Music Player	melodies on the PC speaker
Calendar	a month with today marked
Quiz	a trivia quiz with scoring
Hacker Mode	a retro intrusion animation
SysInfo+	a deeper system information panel
Themes Extra	ten more colour themes
H Language	TechHaven H / H++ / H#, the HavenCode toolchain

```
kyan@havendos:~# bpkg list
bpkg - HavenDOS Package Manager v0.8.0
Available packages:

  fortune          1.0.0  Random quotes and jokes  [installed]
  ascii-art        1.0.0  Big ASCII banner text renderer  [installed]
  paint            1.0.0  ASCII art canvas, save to file  [installed]
  music            1.0.0  PC speaker melodies  [installed]
  calendar         1.0.0  Monthly calendar with today marker
  quiz             1.0.0  Trivia quiz game with scoring
  hacker           1.0.0  Retro hacking terminal animation
  sysinfo-plus     1.0.0  Deep system information panel
  themes-extra     1.0.0  10 additional colour themes
  h-lang           0.2.0  TechHaven H / H++ / H# lang (HavenCode)

Usage: bpkg install <id>  |  bpkg remove <id>  |  bpkg update
       bpkg server <url>  |  bpkg info <id>
kyan@havendos:~#
```

FIGURE 13-9 The package registry.



FIGURE 13-10 An important system utility.



FIGURE 13-11 Large letters.

The screensaver



FIGURE 13-12 Matrix.

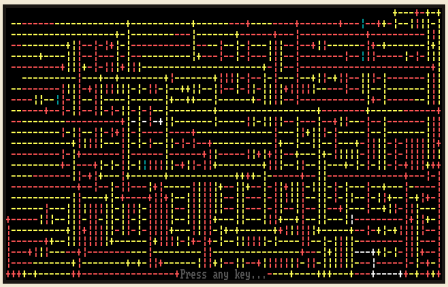


FIGURE 13-13 Pipes.

After two idle minutes a screensaver starts. `screensaver <seconds>` changes the delay and `screensaver off` disables it.

14 Updating, and what to do when it fails

`update` reads a newer ISO from the CD-ROM drive, compares its version with the running one and replaces the kernel on disk. `rollback` puts the previous kernel back. The version comparison is a real one — it will not install an older kernel over a newer one by accident.

> **CAUTION**
>
> `update` needs the CD-ROM to be on the legacy IDE ports. On a Q35 or ICH9 machine it is behind AHCI, which this kernel does not drive, and the update will fail to find the ISO. The boot log says so, at boot, in the `ATA: nothing on 0x1F0/0x170` line.

> **CAUTION**
>
> `install` on a system that is already running HavenDOS will tell you it cannot proceed, and will not touch the disk. The kernel an installed system runs carries no image of itself to write — see section 15. Install from the ISO, or from its live shell.

Working around it

- Switch the virtual machine to an i440FX machine type and attach the CD-ROM as IDE. This is the configuration HavenDOS is tested on.
- Or boot the new ISO and install over the top. The installer rewrites the kernel and bootloader; it does not preserve files on the partition, so copy anything you care about off first.

Checking what you are running

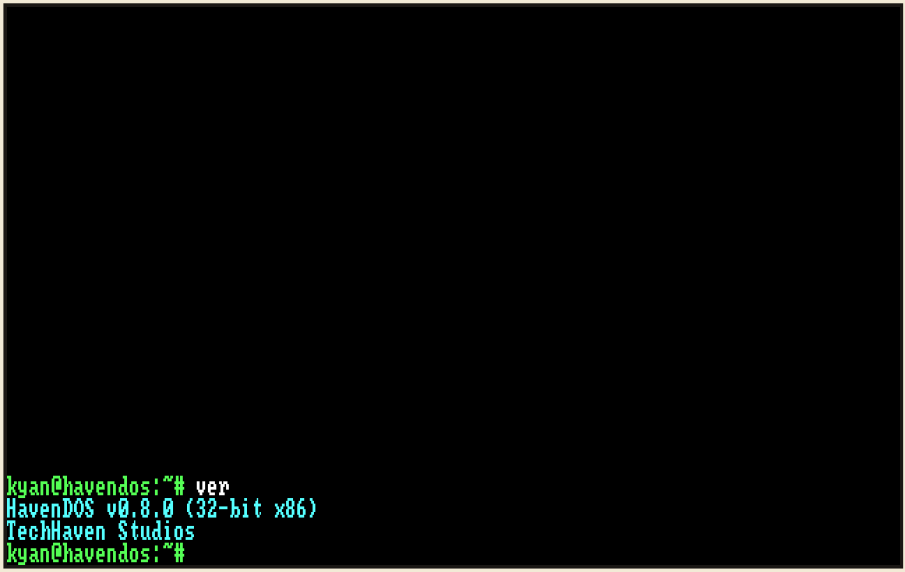


FIGURE 14-1 Version, channel and codename.

The version lives in exactly one header and everything else is generated from it — the kernel, the boot menu titles, the version marker on the installed partition, the splash and the theme label in the GRUB footer. The build refuses to finish if any of them disagree.

15 What this beta cannot do

This section exists because a beta that only lists its features will waste your time. These are the real limits.

A USB stick is seen, not read

Enumeration works, transfers work, and there is no mass-storage class driver on top of them yet. You cannot copy a file off a stick.

Bootable is not the same as usable on modern hardware

The ISO will boot a current UEFI laptop. On that laptop the keyboard is USB behind an xHCI controller, which this kernel does not drive, and the disk is AHCI or NVMe, which it also does not drive. It will come up and you will not be able to type into it. A virtual machine configured as described in section 2 is the supported way to run HavenDOS today.

An installed system cannot install HavenDOS

The build is two-pass: the first pass produces a kernel with no embedded image, the second embeds that kernel and is what ships on the ISO. So the kernel the installer writes to a disk — and therefore the one an installed system runs — has nothing to install. The installer checks before it starts and refuses, rather than formatting a disk and failing at the last step, which is what it used to do. Install from the ISO.

Only one ATA disk is seen

The driver probes the primary master, so a second IDE disk does not appear in the installer's list.

update needs legacy IDE

As in section 14. The diagnosis is in the boot log; the workaround is a machine type change or a reinstall.

Programs still lay out for 80×25

The console can do 80×50 and the framebuffer could do more, but every app draws for an 80×25 grid. They are cleared and redrawn so nothing bleeds through, but they do not yet use a larger one.

The ISO is 16 MB

Earlier releases were five or six. The difference is the UEFI boot image, which is carried alongside the BIOS one so that a single file boots both kinds of machine.

No sound beyond the speaker

There is a PC speaker driver and nothing else. AC’97 is planned, and is what the USB and storage work is ultimately heading towards.

16 Colophon

HavenDOS is written by TechHaven Studios. It began as an experiment in whether a usable system could be written from the first instruction, on a phone, in an emulator. The installer that once only worked against a VirtIO disk now installs from an IDE CD-ROM onto an IDE disk, boots on BIOS and UEFI alike, and enumerates USB devices on chipsets that did not exist when the project started.

Release	0.8.0 Beta, codename “Beacon”
Supersedes	0.7.21, 0.7.22 and 0.7.23, which were one body of work under three numbers
Checks passing	437, up from 303 at 0.7.20
Image	HavenDOS-0.8.0.iso, hybrid BIOS + UEFI, about 16 MB
Part number	THS-HD-0800-OM-A

Set in Courier Prime, with IBM Plex Mono for headings, VT323 for screen listings and Press Start 2P on the cover. Screens were photographed from a running system, not reconstructed.